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Roll No.....

VIIITH SEMESTER

END SEMESTEREXAMINATION

B.Tech.(Computer Engineering)

(November 18)

CO-401

Paper Code

ARTIFICIAL INTELLIGENCE

Time: 3 Hours

Max. Marks: 70

Note: Attempt total 5 questions. Assume missing data if any

Q1. (i) Define and explain the terms: heuristic Search, knowledge.

(ii) Explain the concepts of state space, production rules and control strategy.

(iii) List five project characteristics to analyze a problem domain

[4,5,5]

Q2. Explain with example following search techniques

(i) Hill Climbing,

(ii) Best first Search

(iii) A* Search

[4,4,6]

Q3. (i) Explain the concept of game tree with suitable example

(ii) With the help of Tic Tae Toe game, show how Min max procedure help in deciding a good move for maximizing player.

[7,7]

Q4. Consider the following sentences:

a. Every child loves dog.
b. All dogs have fur skin or short hair.
c. Roby is a dog
d. Tina does not love dog with short hair
e. Shikha loves Roby
f. Tina does not love dog

(i) Represent theses sentences as wff.

(ii) Using backward chaining prove that Tina is not child.

(iii) Convert them to clause form indicating the law used.

(iv) Explaining steps of resolution, prove that Roby has fur skin.

[3,3,4,4]

Q5. (1) Explain the different concepts used in Semantic Net using suitable example. Also write the inference procedure

(i) How semantic nets are extended to semantic Frames. What are its merit and demerits over semantic nets.

[7,7]

Q6. (i)) Draw the architecture of Expert system defining functionality of different components

(ii) Develop logic for representation using fuzzy set with an example of car speed which can be slow, fast and super fast

[10,4]

Q. 7. Write short notes on any two:

(a) Default reasoning or statistical reasoning.

(b) Learning using Neural Network or Genetic algorithm.

(c) AO* Search technique

(d) Crypt arithmetic problems

[7,7]